Design Report

Due: 2025-11-29 at 2200 **Weight:** 20% **Submitter:** Team

Assignment overview

The design report represents your design work in its entirety to a reader unfamiliar with your opportunity framing and design work. So, the design report documents both the design concept you are recommending, and the key parts of the process to arrive at that concept, with an emphasis on your recommendation. The design report answers the question of whether the current design is a verifiable and verified solution to the opportunity presented in the design brief. More specifically, the report aims to answer three key questions:

1. What your recommended design concept, and why is it best current design concept? The word “best” implies comparison, so you should be able to justify the current concept relative to other concepts (and reference designs).
2. What key decisions have determined the design? As you make specific decisions, you need to be sure you understand why you made those decisions and establish the research, testing, or conceptual logic to justify your work.
3. Has the current design concept met sufficient testing to be justifiable as a solution? This question requires you to be able to document your requirements, how they were met, and the kinds of testing you have done.

To accomplish this your team should document your opportunity as reframed, key alternatives from your diverging process, the evaluation of the alternatives using appropriate research and testing, and the recommended design you converged on, though not necessarily in that order.

The design report as a genre in engineering writing does not normally assume the existence of an associated design brief, so it should be written to stand alone. For this assignment, your team will have modified the framing from a design brief, and you should focus your design report on the final framing your team used, rather than on the process of reframing from the design brief.

Assignment stakeholders

This assignment has the following stakeholders:

- Your team, who must report on not only your recommended design concept, but also justify it using the process by which that concept was developed and tested.
- You, individually, as an aspiring engineering designer who must develop both individual and team engineering design skills.
- Your assessor, **who is unfamiliar with your design work to date**, including the design brief, and Alpha, and who is responsible both for assessing and evaluating the engineering design work that your team has completed.
- The stakeholders from your design brief, who by definition have an interest in both the recommended design concept and the process of its development.
- Stakeholders **who may iterate on or implement your recommendation** (e.g. other engineering designers, technicians, contractors, etc.) who want to understand your recommended design concept, its key features, how it was verified, and the results of that verification.

Assignment expectations

Language in this and following sections are to be interpreted as described in RFC 2119.

Objectives

1. Represent and communicate a design opportunity using engineering framing, including an effective set of requirements and evaluation criteria, such that the credibility and quality of candidate design concepts can be assessed.
2. Represent and communicate design concepts and their key features.
3. Demonstrate the verification of design concepts by key requirements, using appropriate engineering prototypes.
4. Demonstrate the use of formal engineering tools to support design work.
5. Demonstrate effective teamwork through a document that reflects a coherent vision of the team's work and a unified voice.[1]
6. Communicate clearly using integrated, structured visual and written representations.

Submission requirements

The design report:

1. **Shall** be submitted as one single PDF file per team through Quercus.
2. **Shall** use a recognized reference format (IEEE, APA, CBE, Chicago, but not MLA).
3. **Should** be typeset in 11-pt serif font with 1.1-line spacing and 1.0-inch margins on standard letter-sized paper.
4. **Shall** not exceed 3000 words, and **should** be approximately 2500 words, including headings, tables, and captions, excluding any title-page content, table of contents, references, and appendices; and **shall** include a word count at the top of the first page.
5. **Shall** include relevant extracts from any used references in an appendix titled "Source Extracts."
6. **Shall** include the original design brief that provided the initial framing for the opportunity in an appendix (Note: there is no need to revise this document. The entire point is simply to see the starting point for the design work and be able to refer to it as needed).

Content requirements

The design report:

1. **Shall** be a standalone document, including appendices, that includes all of the information necessary for assessment and evaluation without the reader needing to locate other documents.
2. Shall document the design opportunity including necessary background information, framing, and the interpreted requirements and evaluation criteria.
3. **Shall** recommend one (1) credible design concept[2] that addresses the framed opportunity.
4. **Shall** document comparisons of the performance of the recommended design concept against at least (≥3) other credible candidate design concepts (which may include reference designs).

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[1] As with all Praxis team assignments, the teaching team assumes that team members share work equitably. In situations where this is not the case, students or instructors should refer to Section 7 of the Syllabus.

[2] If there is good reason, this may include the null option, which is to recommend no change from the status quo, or an existing design, if credible design work overall reveals that there is an existing design that resolves the opportunity, even after thorough and credible reframing/scoping. Both of these options should only be considered after thorough design work to generate new concepts, as they require different and much more substantial justification.

5. **Shall** integrate a Pugh chart
   (or similar representation that includes all of the content present in a Pugh chart).
6. **Shall** justify three (3) key design decisions in the recommended design, using a combination of verification (testing), prototypes, research, and or calculation.
7. **Shall** include up to three (≤3) minutes of video content, linked in the relevant section of the design report, to represent a least one (≥1) key aspect of your design work (i.e., a key feature/working principle of (a) design concept(s), the verification of (a) design concept(s), etc.).
   - This video will ideally be uploaded to an online location[3] with a link provided in the relevant part of the design report. It is your team's responsibility for ensuring your assessors will have access to the video when they click on the link. Alternatively, you may choose to submit a video file together with your design report. The maximum allowed video file size is 100 MB **in total**. Any uploaded video submissions with a total file size larger than 100 MB will not be assessed. If you host the video online yourself, we have no opinions about file size.
8. **Shall** document using at least one (≥1) prototype in verification activities.

Evaluation criteria

These criteria describe the attributes that will be evaluated. For the full metrics consult the Design Report Independent Assessment Tool. That tool places these criteria alongside descriptions of levels of achievement that you can use to independently assess your work. For all criteria, better is considered to be "more", "higher", or "greater".

Note: These criteria may not be weighted equally.

1. Quality and credibility of engineering argument, with emphasis on justifying claims through credible use of engineering-appropriate sources (including, but not limited to, research or testing)
2. Quality of the framing, including background, engineering requirements, and evaluation criteria.
3. Quality of the application of engineering tools (e.g. for: exploring (diverging); assessing (converging); representing (e.g. prototyping); and, verifying).
4. Quality of the recommended concept, as demonstrated through verification against the requirements and assessment using evaluation criteria.
5. Quality of the verification process of the recommended design concept as represented by appropriate prototypes that were used in testing.
6. Quality of the document as a design report.
7. Coherence and clarity of written, visual, and video communication.
8. Coherence of the vision the team's work represented in the writing style of the document.

[3] As a U of T student, you have access the MyMedia, which can host video content online. https://mymedia.library.utoronto.ca/login